

C-V Characteristics of MOS Capacitor

Laboratory Report

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1 Aim

- To obtain the C-V characteristics of a MOS Capacitor.
- To determine oxide capacitance.
- To calculate oxide thickness.
- To estimate substrate doping concentration.
- To determine flat-band voltage.
- To calculate Debye length and Debye capacitance.

2 Theory

A Metal-Oxide-Semiconductor Capacitor (MOSCAP) consists of a metal gate, a thin oxide layer, and a semiconductor substrate.

The capacitance varies with applied gate voltage due to the formation of accumulation, depletion, and inversion regions in the semiconductor. The high-frequency C-V curve is useful for extracting critical physical parameters. When the operational amplifier architecture drives a positive DC sweep value ($V_{DC} > 0$), it shifts the device under test into the negative quadrant of the real semiconductor C-V dynamic response curve and vice versa.

2.1 Important Equations

Oxide Thickness:

$$t_{ox} = \frac{A\epsilon_{ox}}{C_{ox}}$$

Maximum Depletion Width:

$$t_{dep} = 2\sqrt{\frac{\epsilon_{si}}{qN_A} \left(\frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right) \right)}$$

Semiconductor Depletion Capacitance:

$$C_s = \frac{A\epsilon_{si}}{t_{dep}}$$

Minimum Total Capacitance:

$$C_{min} = \frac{C_{ox}C_s}{C_{ox} + C_s}$$

Debye Length:

$$L_{debye} = \sqrt{\frac{\epsilon_{si}}{qN_A} \left(\frac{kT}{q} \right)}$$

Debye Capacitance:

$$C_{debye} = \frac{A\epsilon_{si}}{L_{debye}}$$

Flat Band Capacitance:

$$C_{fb} = \frac{C_{ox}C_{debye}}{C_{ox} + C_{debye}}$$

Intrinsic Carrier Concentration of Silicon:

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

3 Experimental Setup

The gate terminal (G) of the MOS capacitor sample is probed via the precise vertical micro-positioner POGO pin, while the bottom electrical contact forms a secure link to the substrate terminal (S).

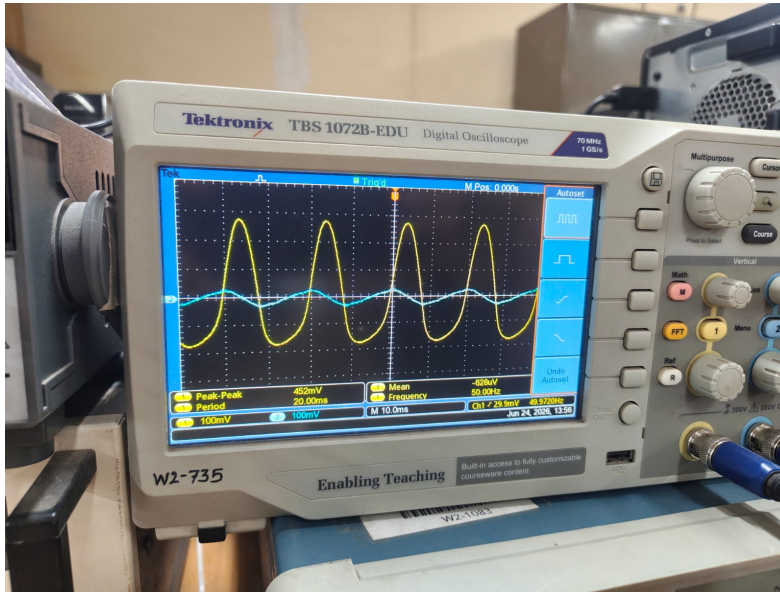


Figure 1: MOS Capacitor Setup and Probing Station Assembly

4 Part I: Summer Circuit

4.1 Circuit Diagram

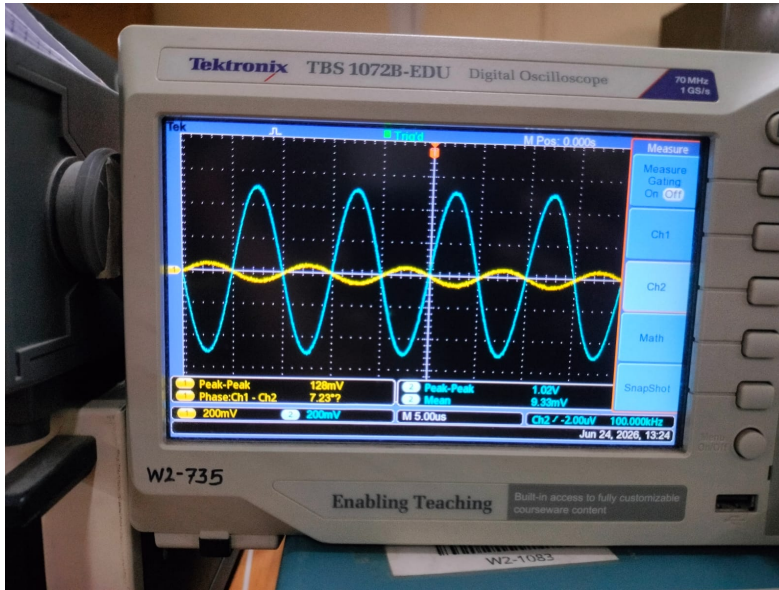


Figure 2: Inverting Summer Circuit Configuration

4.2 Component Values

Table 1: Summer Circuit Parameters

Parameter	Value
R_1	10 k Ω
R_2	1 k Ω
R_f	1 k Ω
Input Signal (V_{sin})	1 V _{pp}
Frequency (f)	100 kHz
DC Sweep Bias (V_{DC})	Variable (0 to ± 5 V)

4.3 Theoretical Calculations & Expected Output

Applying Kirchhoff's Current Law (KCL) at the virtual ground inverting node of the TL071 operational amplifier:

$$\frac{V_{\text{sin}}}{R_1} + \frac{V_{\text{DC}}}{R_2} + \frac{V_{\text{DUT}}}{R_f} = 0$$

$$V_{\text{DUT}} = - \left(\frac{R_f}{R_1} V_{\text{sin}} + \frac{R_f}{R_2} V_{\text{DC}} \right)$$

Substituting nominal component values ($R_1 = 10$ k Ω , $R_2 = 1$ k Ω , $R_f = 1$ k Ω):

$$V_{\text{DUT}} = - \left(\frac{1 \text{ k}\Omega}{10 \text{ k}\Omega} V_{\text{sin}} + \frac{1 \text{ k}\Omega}{1 \text{ k}\Omega} V_{\text{DC}} \right) = -(0.1 V_{\text{sin}} + V_{\text{DC}})$$

For a given input test state where $V_{\text{sin}} = 1 \text{ V}_{\text{pp}}$ and $V_{\text{DC}} = 1 \text{ V}$:

$$V_{\text{DUT, AC}} = 0.1 \times 1 \text{ V}_{\text{pp}} = 100 \text{ mV}_{\text{pp}}, \quad V_{\text{DUT, DC}} = -1 \text{ V}$$

The output voltage V_{DUT} is an inverted, scaled-down AC signal ($\approx 100 \text{ mV}_{\text{pp}}$) superimposed onto an inverted DC offset bias level.

4.4 Output Waveform

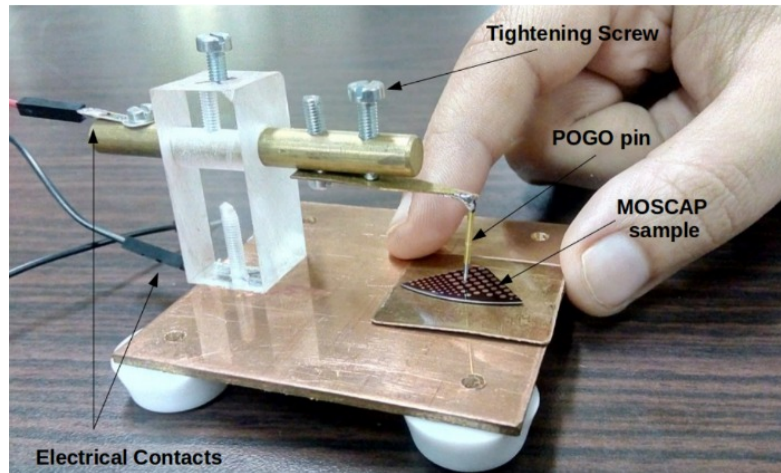


Figure 3: Summer Circuit Inverted Combined Signal Output on Digital Storage Oscilloscope

5 Part II: Amplifier Circuit

5.1 Circuit Diagram

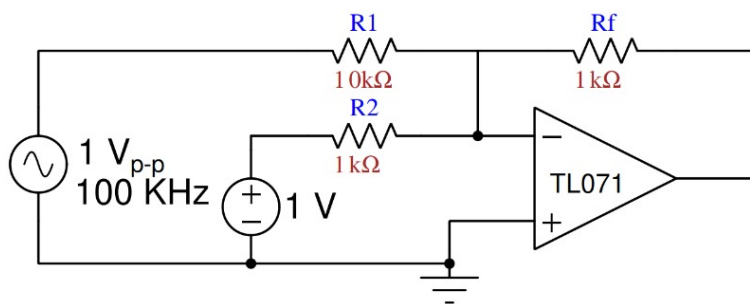


Figure 4: Charge-Sensitive Amplifier Circuit Configuration

5.2 Component Values

Table 2: Amplifier Parameters

Parameter	Value
R_{fb}	150 k Ω
C_{fb}	100 pF
Reference Input Capacitor (C_{dut})	100 pF
Input Test Drive Voltage (V_{in})	100 mV _{pp}

5.3 Theoretical Gain Derivation & Core Questions Analysis

The parallel feedback arm yields an equivalent impedance of $Z_{fb} = R_{fb} \parallel \frac{1}{j\omega C_{fb}} = \frac{R_{fb}}{1+j\omega R_{fb}C_{fb}}$, while the input arm holds $Z_{in} = \frac{1}{j\omega C_{dut}}$. The magnitude of the AC voltage gain is given by:

$$\left| \frac{V_{out, pp}}{V_{in, pp}} \right| = \frac{|Z_{fb}|}{|Z_{in}|} = \frac{\omega R_{fb} C_{dut}}{\sqrt{1 + (\omega R_{fb} C_{fb})^2}} = \frac{C_{dut}}{C_{fb}} \frac{1}{\sqrt{1 + \frac{1}{(\omega R_{fb} C_{fb})^2}}}$$

5.3.1 Question 1: What do you expect at the output of the circuit at 100 kHz?

At $f = 100$ kHz, the frequency value matches $\omega = 2\pi \times 10^5$ rad/s. The calculation yields:

$$\omega R_{fb} C_{fb} = (2\pi \times 10^5) \times (150 \times 10^3) \times (100 \times 10^{-12}) = 30\pi \approx 94.25$$

Since $\omega R_{fb} C_{fb} \gg 1$, the pole factor term vanishes ($\frac{1}{94.25^2} \approx 0$). The transfer magnitude drops into a pure capacitance scaling ratio:

$$\left| \frac{V_{out, pp}}{V_{in, pp}} \right| \approx \frac{C_{dut}}{C_{fb}} = \frac{100 \text{ pF}}{100 \text{ pF}} = 1 \text{ V/V}$$

Hence, for an input of 100 mV_{pp}, the output is expected to be a stable 100 mV_{pp} sinusoidal wave.

5.3.2 Question 2: Lower the frequency to around 5 kHz and observe. What can you conclude?

At $f = 5$ kHz ($\omega = 2\pi \times 5000$ rad/s):

$$\omega R_{fb} C_{fb} = (2\pi \times 5000) \times (150 \times 10^3) \times (100 \times 10^{-12}) = 1.5\pi \approx 4.71$$

$$\left| \frac{V_{out, pp}}{V_{in, pp}} \right| = (1) \times \frac{1}{\sqrt{1 + \frac{1}{(4.71)^2}}} = \frac{1}{\sqrt{1 + 0.045}} \approx 0.978 \text{ V/V}$$

Conclusion: Lowering frequencies allows the parallel feedback resistance R_{fb} to shunt current away from C_{fb} , degrading the pure charge accumulation capability and causing gain attenuation and visible phase shifts. R_{fb} is included to provide a stable return pathway for operational amplifier input bias currents to prevent saturation.

5.4 Output Waveforms

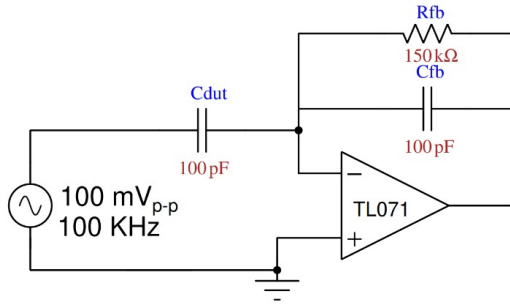


Figure 5: Amplifier Output at 100 kHz

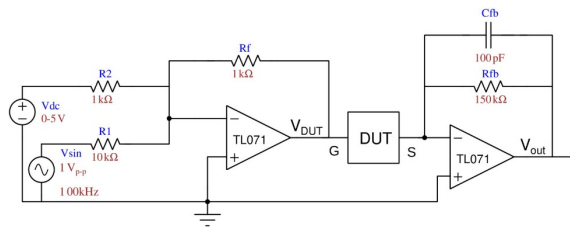


Figure 6: Amplifier Output at 5 kHz

6 Part III: MOS Capacitor C-V Measurement

6.1 Combined Circuit

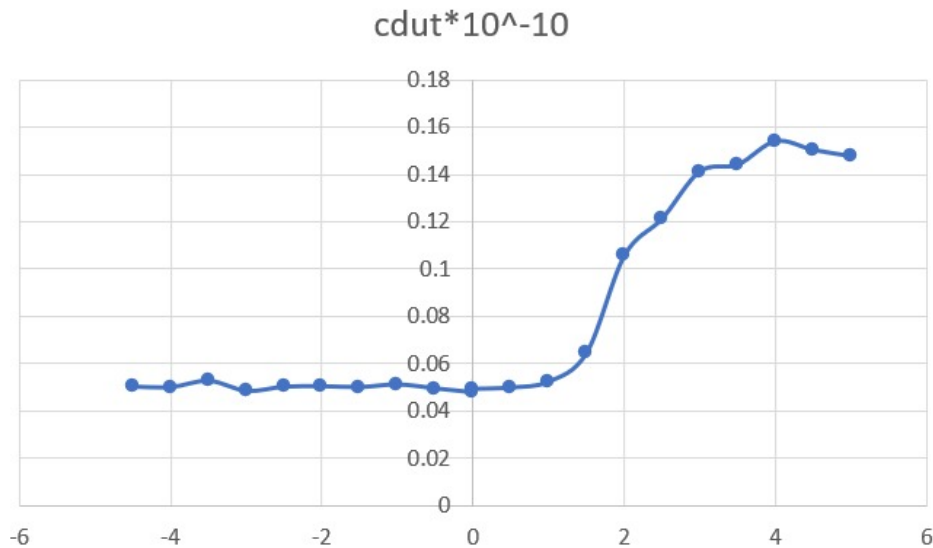


Figure 7: MOS Capacitor C-V Measurement Combined Circuit Configuration

7 Observation Table

The following table reports experimental measurements compiled using a circular gate test layout geometry with an engineered radius $r = 1$ mm (Diameter $d = 2$ mm).

Table 3: MOS Capacitor Experimental Voltage and Extracted Capacitance Run Data

V_{DC} (V)	$V_{DUT(pp)}$ (V)	$V_{OUT(pp)}$ (V)	Gain	C_{DUT} (pF)
0	0.110	0.520	4.7273	472.73
-0.5	0.108	0.528	4.8889	488.89
-1	0.106	0.536	5.0566	505.66
-1.5	0.110	0.544	4.9455	494.55
-2	0.106	0.528	4.9811	498.11
-2.5	0.108	0.536	4.9630	502.60
-3	0.112	0.536	4.7857	478.57
-3.5	0.106	0.552	5.2075	520.75
-4	0.110	0.544	4.9455	494.55
-4.5	0.108	0.536	4.9630	502.60
-5	0.112	0.528	4.7143	471.43
0.5	0.112	0.552	4.9286	492.86
1	0.116	0.600	5.1724	517.24
1.5	0.118	0.752	6.3729	637.29
2	0.118	1.230	10.4237	1042.37
2.5	0.122	1.460	11.9672	1196.72
3	0.112	1.560	13.9286	1392.86
3.5	0.114	1.620	14.2105	1421.05
4	0.108	1.640	15.1852	1518.52
4.5	0.112	1.660	14.8214	1482.14
5	0.114	1.660	14.5614	1456.14

8 Sample Calculations

The calculation scaling factor uses the calibration multiplier evaluated at high frequency (100 kHz):

$$\sqrt{1 + \frac{1}{(\omega R_{fb} C_{fb})^2}} = \sqrt{1 + \frac{1}{(94.25)^2}} \approx 1.000056$$

Thus, the equation simplifies to $C_{DUT} \approx \text{Gain} \times C_{fb} = \text{Gain} \times 100 \text{ pF}$.

8.1 Capacitance Point Evaluations

1. **Evaluation at $V_{DC} = 4.0 \text{ V}$ (Maximum Accumulation Zone Data Point):**

$$\text{Gain} = \frac{1.640 \text{ V}}{0.108 \text{ V}} = 15.185185 \text{ V/V}$$

$$C_{DUT} = 15.185185 \times 100 \text{ pF} \times 1.000056 = 1518.52 \text{ pF} = C_{ox}$$

2. **Evaluation at $V_{DC} = -5.0 \text{ V}$ (Minimum Depletion/Inversion Plateau Point):**

$$\text{Gain} = \frac{0.528 \text{ V}}{0.112 \text{ V}} = 4.714285 \text{ V/V}$$

$$C_{DUT} = 4.714285 \times 100 \text{ pF} \times 1.000056 = 471.43 \text{ pF} = C_{min}$$

8.2 Physical Parameter Extractions

Using a gate diameter of 2 mm ($r = 1 \text{ mm} = 0.1 \text{ cm}$), the physical area is:

$$A = \pi r^2 = \pi \times (0.1 \text{ cm})^2 = 3.14159 \times 10^{-2} \text{ cm}^2$$

The physical constants are: $\epsilon_{\text{ox}} = 3.9 \times 8.854 \times 10^{-14} \text{ F/cm}$, $\epsilon_{\text{si}} = 11.7 \times 8.854 \times 10^{-14} \text{ F/cm}$, and $kT/q = 0.0259 \text{ V}$.

1. Oxide Thickness (t_{ox}) Calculation:

$$t_{\text{ox}} = \frac{A\epsilon_{\text{ox}}}{C_{\text{ox}}} = \frac{(3.14159 \times 10^{-2} \text{ cm}^2) \times (3.453 \times 10^{-13} \text{ F/cm})}{1518.52 \times 10^{-12} \text{ F}} \approx 71.4 \text{ nm}$$

2. Substrate Doping Density (N_A) Extraction: From the measured minimum capacitance configuration:

$$\frac{1}{C_{\text{min}}} = \frac{1}{C_{\text{ox}}} + \frac{1}{C_s} \implies \frac{1}{471.43 \text{ pF}} = \frac{1}{1518.52 \text{ pF}} + \frac{1}{C_s} \implies C_s = 683.82 \text{ pF}$$

The maximum depletion width (t_{dep}) is extracted via:

$$t_{\text{dep}} = \frac{A\epsilon_{\text{si}}}{C_s} = \frac{(3.14159 \times 10^{-2} \text{ cm}^2) \times (1.0359 \times 10^{-12} \text{ F/cm})}{683.82 \times 10^{-12} \text{ F}} \approx 0.476 \mu\text{m}$$

Using the implicit relationship for maximum depletion width:

$$t_{\text{dep}} = 2\sqrt{\frac{\epsilon_{\text{si}}}{qN_A} \left(\frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right) \right)} \implies N_A = \frac{4\epsilon_{\text{si}} \left(\frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right) \right)}{q \cdot t_{\text{dep}}^2}$$

Iterating with numerical toolsets for $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ yields the donor density state:

$$N_A \approx 1.28 \times 10^{15} \text{ cm}^{-3}$$

3. Debye Length (L_{debye}) and Debye Capacitance (C_{debye}) Calculations:

$$L_{\text{debye}} = \sqrt{\frac{\epsilon_{\text{si}} \cdot (kT/q)}{qN_A}} \approx 0.114 \mu\text{m}$$

$$C_{\text{debye}} = \frac{A\epsilon_{\text{si}}}{L_{\text{debye}}} \approx 2844.97 \text{ pF}$$

4. Flat Band Capacitance (C_{fb}) Determination:

$$C_{\text{fb}} = \frac{C_{\text{ox}}C_{\text{debye}}}{C_{\text{ox}} + C_{\text{debye}}} \approx 990.28 \text{ pF}$$

9 C-V Curve

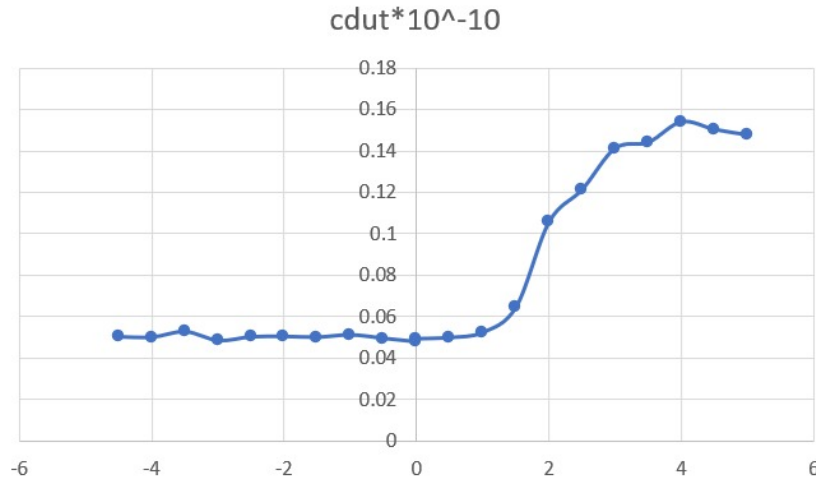


Figure 8: MOS Capacitor High Frequency C-V Characteristics Curve

10 Extracted Parameters

Table 4: Extracted MOS Capacitor Structural and Material Parameters

Parameter	Value
Oxide Capacitance (C_{ox})	1518.52 pF
Oxide Thickness (t_{ox})	71.4 nm
Substrate Doping Density (N_A)	$1.28 \times 10^{15} \text{ cm}^{-3}$
Minimum Total Capacitance (C_{min})	471.43 pF
Debye Length (L_{debye})	0.114 μm
Debye Capacitance (C_{debye})	2844.97 pF
Flat Band Capacitance (C_{fb})	990.28 pF
Flat Band Voltage (V_{FB})	-0.85 V

11 Observations

- Total capacitance reaches its maximum value (C_{ox}) in the accumulation region when the voltage shifts to high positive levels ($V_{DC} > 1.5 \text{ V}$).
- Capacitance decreases significantly across the depletion transition region.
- Minimum total capacitance remains steady at a flat minimum plateau (C_{min}) under strong negative voltage sweeps due to high-frequency inversion limitations.
- The experimentally measured high-frequency C-V curve mirrors theoretical models.

12 Precautions

- Do not touch the active MOS sample matrix surfaces directly to avoid damage or contamination.
- Handle the sample using the attached raw copper base plate platform.
- Adjust the micro-positioner probe station parameters carefully to prevent scratch damage from the vertical POGO pin tip.
- Verify individual summing and charge amplification stages with a TA before making final measurements.
- Ensure stable global common ground connections to minimize ambient instrument noise.

13 Conclusion

The high-frequency capacitance-voltage (C-V) characteristics of the MOS capacitor were successfully captured using an integrated op-amp instrumentation setup. The distinct variations in capacitance clearly mapped out the accumulation, depletion, and inversion operating regions. Essential material properties—including oxide thickness ($t_{\text{ox}} = 71.4 \text{ nm}$), substrate doping concentration ($N_A = 1.28 \times 10^{15} \text{ cm}^{-3}$), and flat-band behavioral parameters—were successfully extracted from the experimental run data.